

EMPIRICAL BAYES ESTIMATION FOR DISCRETE DISTRIBUTIONS

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Summary: A number of families of discrete distributions are considered, and Bayes estimators for their parameters are investigated. Assuming the usual squared error loss, this means that the posterior expectation is the quantity of interest. But this implicitly involves the prior distribution on the parameter, which in certain cases it is not necessary to directly find. The consequent simple empirical Bayes estimators are the subject of this article. It is found that in all but two of the discrete distributions under study such estimators are derivable, and where they are not, approximations are available.

1. INTRODUCTION

Counts are the basic quantities of data collection and data analysis. Even so-called continuous variables must finally be observed at discrete levels, or possibly they are grouped into discrete classes for analysis and inference purposes. For this reason probability distributions on countable sets, henceforth the set of non-negative integers, offer a great richness for a would-be modeller of real-world phenomena. Immediately the binomial, hypergeometric, Poisson distributions

come to mind, however less known families such as those of Sichel (1975) have proved themselves in the "market-place", especially for phenomena (e.g. word counts) which exhibit a longer-tailedness than the usual discrete distributions are capable of modelling.

Typically the problem facing the statistician is the following. He has collected observations $X_1, \dots, X_N$ from some probability distribution which he has modelled as $h(x|\xi)$. The parameter ξ is interpretable with regard to the phenomena under study, and so some inference is needed from data $(X_1, \dots, X_N)$ to parameter ξ .

Sometimes however, this interpretation that ξ is a *fixed* but unknown parameter, is unrealistic. For example, latent trait models in mental testing (Lord and Novick, 1968) must realistically model a sequence of bivariate random vectors $(X_1, Z_1), \dots, (X_N, Z_N)$, where Z_i is the *unobserved* trait (or *true score*) of the i -th examinee, and X_i is that examinee's *observed score* on a test. The population from which the examinee comes has a certain distribution of true scores, call it G . So if Z is the true score of a randomly chosen member of that population, then $G(\xi) = \Pr\{Z \leq \xi\}$. Thus there are contexts where it makes sense to model the unknown parameter ξ as also being random. A sensible question then to ask is: Suppose I observe an $X = x$, what can I then say about the associated (unobserved) Z ?

The approach we have been describing in the paragraph above would be called empirical Bayes (see Robbins, 1956, 1964, and Maritz, 1970) if we used other observed variables $X_1, \dots, X_N$ to make an inference from observed $X = x$ to unobserved Z . Section 2 briefly sets out the general ideas behind the empirical Bayes approach and considers the very important notion of a *simple* empirical Bayes estimator. Up to now these have been discovered in a very haphazard way, however an identity by Cressie (1982)

shows their appearance not to be happenstance. Maritz and Lwin (1975) derived something similar for continuous variables; in this section no specific assumption of continuous or discrete is made. It is only in section 3 that the methods of section 2 are applied to a comprehensive list of *discrete* variables. Section 4 contains some concluding remarks.

2. THE EMPIRICAL BAYES APPROACH

2.1 The general model

Consider a probability space generated by the bivariate random vector (X, Z) , where the first member is observable but the second is not. Let Z have an unknown distribution function G , and given $Z = \xi$, X is modelled to have probability distribution or density $h(x|\xi)$; h known. Hence the marginal distribution of X is

$$(2.1) \quad \phi_G(x) = \int h(x|\xi) dG(\xi) .$$

2.2 The useful identity

Suppose L is a *linear* functional with domain contained in the space of all real valued functions. Then provided the range of Z does not depend upon x ,

$$(2.2) \quad L(\phi_G(\cdot)) = \int L(h(\cdot|\xi)) dG(\xi) .$$

Define the functional

$$(2.3) \quad R: f(\cdot) \rightarrow L(f(\cdot))/f(\cdot) ,$$

where f is any real valued function. Cressie (1982) has shown

$$(2.4) \quad R(\phi_G(x)) = \int R(h(x|\xi)) dP(\xi|x) ,$$

where $P(\xi|x)$ is the posterior distribution function of Z , given $X = x$.

2.3 Minimizing squared error loss

Given $X = x$, we wish to make a decision $\delta(x)$ about the associated value of Z . Let a loss $L(\delta(x), \xi) = (\delta(x) - \xi)^2$ be incurred when the parameter value is ξ and the decision $\delta(x)$ is made. The overall expected loss is then $E(\delta(X) - Z)^2$, and the resulting δ which minimizes this is called the *Bayes estimator*. It is just the posterior mean

$$(2.5) \quad \begin{aligned} \delta_G(x) &= \int \xi dP(\xi|x) \\ &= (\phi_G(x))^{-1} \int \xi h(x|\xi) dG(\xi) . \end{aligned}$$

If G was known, the decision δ_G , a function of observed $X = x$, would be easy to calculate. The "empirical" part of empirical Bayes involves using an already observed sample $X_1, \dots, X_N$ (N independent and identically distributed observations from ϕ_G) to estimate δ_G and possibly G .

2.4 Simple empirical Bayes estimators

An empirical Bayes estimator $\hat{\delta}_G(x)$ will be called *simple* if its corresponding Bayes estimator $\delta_G(x)$ can be written as a functional only of ϕ_G and x . For example, suppose

$$h(x|\xi) = h_p(x|\xi) = e^{-\xi} \xi^x / x!; \quad x = 0, 1, 2, \dots,$$

the Poisson distribution. Then it is easy to check that

$$\delta_G(x) = E(Z|X=x) = (x+1)\phi_G(x+1)/\phi_G(x) .$$

Obviously simple empirical Bayes estimators are most desirable, since the difficulty of first estimating and/or modelling G has been finessed. No form for G is assumed; the only place where the statistician is expected to (and indeed he would want to) model is in the choice of $h(x|\xi)$.

2.5 The identity and simple empirical Bayes

The development in subsection 2.2 was for *any* linear functional L , and hence a wide choice of functionals R (see (2.3)) is available to apply to (2.4). Thus if an R_0 can be found for which,

$$(2.6) \quad R_0(h(x|\xi)) = \xi u(x) + v(x) ,$$

where $u > 0$, v are real valued functions, then substitution of this R_0 into (2.4) yields

$$(2.7) \quad \delta_G(x) = \{R_0(\phi_G(x)) - v(x)\}/u(x) ,$$

which depends *only* on ϕ_G and x . Therefore an observed sample $X_1, \dots, X_N$ can be used to estimate ϕ_G , and by implication $\delta_G(x)$. The result: a *simple empirical Bayes estimator* of Z , given $X=x$.

We will show in the next section that for very many of the one parameter discrete distributions, a simple empirical Bayes estimator is available. Sometimes a reparameterization is necessary, which is not serious when interpretations are possible (as they often are) on a different scale. But even if the original parameterization is insisted upon, techniques such as those described in Cressie (1982) for the binomial distribution give an approximate simple empirical Bayes estimator

on the original scale. Much of current research in empirical Bayes is involved with efficient estimation of ϕ_G in (2.7). We will touch only briefly on this (section 4), considering a large part of the problem solved when a Bayes estimator of the form (2.7) is found.

3. DISCRETE DISTRIBUTIONS

Our general approach in this section is to give the Bayes estimator (2.7), which would yield a simple empirical Bayes estimator for a discrete h . Most of the distributions h defined below can be found in Johnson and Kotz (1969); less well known distributions will be given specific references. We do allow a possible reparameterization first.

3.1 Binomial distribution, parameter $0 < p < 1$

$$h_b(x|p) = \binom{k}{x} p^x (1-p)^{k-x}; \quad x = 0, 1, \dots, k.$$

Put $\xi = p/(1-p)$,

$$R_0\{f(x)\} = f(x+1)/f(x), \quad u(x) = (k-x)/(x+1), \quad v(x) = 0.$$

Then from (2.7),

$$(3.1) \quad \delta_G(x) = \{(x+1)\phi_G(x+1)\}/\{(k-x)\phi_G(x)\}; \quad x = 0, 1, \dots, k-1.$$

For more details on this interesting case, and for what to do when $x = k$, we refer the reader to Cressie (1982).

3.2 Hypergeometric distribution, integer parameter $0 \leq d \leq M$

$$h_h(x|d) = \frac{\binom{d}{x} \binom{M-d}{m-x}}{\binom{M}{m}}; \quad x = 0 \vee d - (M-m), \dots, m \wedge d,$$

where $a \vee b \equiv \max(a, b)$, $a \wedge b \equiv \min(a, b)$.

Now the set $\{(x, d): 0 \vee d - (M-m) \leq x \leq m \wedge d\} = \{(x, d): x \leq d \leq x + (M-m) \text{ and } 0 \leq x \leq m\}$. Hence

$$(3.2) \quad \phi_G(x) = \sum_{d=x}^{x+(M-m)} h_h(x|d)g(d); \quad 0 \leq x \leq m.$$

However the identity of subsection 2.2 cannot be applied directly because the support of the prior $g(\cdot)$ depends on x .

Usually d has the interpretation as the number of defectives in a large batch of size M ; a sample of size m is taken (without replacement) and h_h is the probability law of the number of defectives in the sample. In practice M is large and since the cost of sampling is a consideration, m is usually small compared to M . Moreover the number of defectives d should only be a small proportion of M . A parts manufacturer for example, would go out of business if d was consistently large. Hence for most applications, it is not unreasonable to assume that

$$(3.3) \quad d < M - m \quad \text{a.s.}$$

The assumption (3.3) means that (3.2) becomes

$$(3.4) \quad \phi_G(x) = \sum_{d=x}^{M-m} h_h(x|d)g(d); \quad 0 \leq x \leq m,$$

which is now manageable for what is to follow. For $0 \leq x \leq m-1$,

$$\begin{aligned}\phi_G(x+1) &= \sum_{d=x+1}^{M-m} h_h(x+1|d)g(d) \\ &= \sum_{d=x+1}^{M-m} \frac{(d-x)}{(x+1)} \frac{(m-x)}{(M-m+x+1-d)} h_h(x|d) g(d); \end{aligned}$$

$$\text{i.e. } \frac{\phi_G(x+1)(x+1)}{\phi_G(x)(m-x)} = \sum_{d=x}^{M-m} \frac{d-x}{M-m+1-(d-x)} p(d|x)$$

The first term of the summand is clearly *not* of the form (2.6) where $\xi = w(d)$, some function of d .

$$\text{However } \frac{(d-x)}{M-m+1-(d-x)} = \frac{d}{M-d} \left\{ \frac{1-x/d}{1-(m-x-1)/(M-d)} \right\},$$

and any reasonable prior will be such that most of the probability in $p(d|x)$ will be where $x/d = m/M$. Hence the term in braces $\simeq 1$.

Put $\xi = d/(M-d)$.

Then from (2.5)

$$(3.5) \quad \delta_G(x) = \{(x+1)\phi_G(x+1)\}/\{(m-x)\phi_G(x)\}; \quad x = 0, 1, \dots, m-1,$$

will yield an *approximate* simple empirical Bayes estimator. Compare this to (3.1); the hypergeometric approximation to the binomial is well known.

3.3 Sichel's distributions

In the search for a family of discrete distributions which would adequately explain word frequency distributions, Sichel (1975)

discovered an extremely flexible three parameter family,

$$(3.6) \quad h(x|\alpha, \theta, \gamma) = \frac{((1-\theta)^{\frac{1}{2}})^{\gamma} (\alpha\theta/2)^x}{K_{\gamma}(\alpha(1-\theta)^{\frac{1}{2}}) x!} K_{x+\gamma}(\alpha); \quad x = 0, 1, 2, \dots,$$

where $\alpha > 0$, $0 < \theta < 1$, $-\infty < \gamma < \infty$, and $K_{\gamma}(\cdot)$ is the modified

Bessel function of the second kind of order γ . The Bessel recurrence formula $K_{\nu+1}(z) = (2\nu/z)K_{\nu}(z) + K_{\nu-1}(z)$ then gives

$$h(x|\alpha, \theta, \gamma) = \{\theta(x+\gamma-1)/x\} h(x-1|\alpha, \theta, \gamma) + \{(\alpha\theta)^2/4x(x-1)\} h(x-2|\alpha, \theta, \gamma).$$

On application of this family to 18 works by authors as diverse as Macaulay, Bunyan, Shakespeare etc., Sichel found the special case $\gamma = -\frac{1}{2}$ to be appropriate, and that estimates of θ were remarkably stable: the 18 $\hat{\theta}$'s have a median of 0.981 and a median absolute deviation (MAD) of 0.009. Hence the only real parameter variation which is occurring in these word frequency distributions is in α . Henceforth consider the one parameter Sichel distribution

$$h_s(x|\alpha) = h(x|\alpha, 0.98, -\frac{1}{2}); \quad x = 0, 1, 2, \dots,$$

where the right-hand side is defined by (3.6). Then the above recurrence relation becomes

$$h_s(x+2|\alpha) = \{0.98(x+\frac{1}{2})/(x+2)\} h_s(x+1|\alpha) + \{0.49^2\alpha^2/(x+2)(x+1)\} h_s(x|\alpha),$$

and hence

$$\frac{h_s(x+2|\alpha)}{h_s(x|\alpha)} - \frac{0.98(x+\frac{1}{2})}{(x+2)} \frac{h_s(x+1|\alpha)}{h_s(x|\alpha)} = \frac{0.49^2\alpha^2}{(x+2)(x+1)}.$$

Put $\xi = \alpha^2$,

$$R_0(f(x)) = [f(x+2) - \{0.98(x+\frac{1}{2})/(x+2)\}f(x+1)]/f(x), \quad u(x) = 0.49/(x+2)(x+1),$$

$$v(x) = 0$$

Then from (2.7) ,

$$(3.7) \quad \delta_G(x) = \frac{(x+2)(x+1)}{0,49^2} \frac{[\phi_G(x+2) - \{0,98(x+\frac{1}{2})/(x+2)\}\phi_G(x+1)]}{\phi_G(x)}; \quad x=0,1,2,\dots$$

3.4 Other discrete distributions

We have chosen three difficult examples to illustrate the method, but we would like to emphasize the breadth of applicability of the empirical Bayes identity (2.7). Table 1 shows six distributions; respectively the geometric distribution h_g , the negative binomial distribution h_n , the Poisson distribution h_p , Fisher's logarithmic series distribution h_f , the Riemann distribution h_r (see Rider, 1965), a distribution by Good (1953) h_1 ; along with the parameter specification and the simple Bayes estimate δ_G .

4. CONCLUDING REMARKS

Most of the discrete distributions considered will yield exact simple empirical Bayes estimators. Only two, the hypergeometric distribution and the Riemann distribution, need an approximation. The useful identity (Cressie, 1982) has proved its worth in the systematic treatment of each of the distributions.

The next stage, which we only briefly discuss here, is the substitution of estimators of $\phi_G(\cdot)$ into (2.7). In most cases it is not enough just to use

$$\hat{\phi}_G(x) = \frac{\#X_1, \dots, X_N = x}{N},$$

since usually $\delta_G(x)$, the posterior expectation, must satisfy

TABLE 1
Simple Bayes estimators of $\xi = \theta$, for various discrete distributions.

DISTRIBUTION	PARAMETER	$\delta_G(x)$
$h_g(x \theta) = \theta^x(1-\theta) ; x = 0,1,2,\dots$	$0 < \theta < 1$	$\phi_G(x+1)/\phi_G(x)$
$h_n(x \theta) = \binom{x+k-1}{x} \theta^x(1-\theta)^k ; x = 0,1,2,\dots$	$0 < \theta < 1$	$\{(x+1)\phi_G(x+1)\}/\{(x+k)\phi_G(x)\}$
$h_p(x \theta) = e^{-\theta} \theta^x / x! ; x = 0,1,2,\dots$	$\theta > 0$	$\{(x+1)\phi_G(x+1)/\phi_G(x)\}$
$h_f(x \theta) = \{-\ln(1-\theta)\}^{-1} \theta^x / x ; x = 1,2,\dots$	$0 < \theta < 1$	$\{(x+1)\phi_G(x+1)/x\phi_G(x)\}$
$* h_r(x \theta) = (\text{zeta}(-\theta))^{-1} x^{-\theta} ; x = 1,2,\dots$	$\theta < -1$	$(x+1)\{\phi_G(x+1)/\phi_G(x) - 1\}$
$h_1(x \theta) = \theta^{x+1} / \{x(x+1)(\theta + (1-\theta)\ln(1-\theta))\} ; x = 1,2,\dots$	$0 < \theta \leq 1$	$(x+2)\phi_G(x+1)/x\phi_G(x)$

* Here $\text{zeta}(\cdot)$ is the Riemann zeta function; also $\delta_G(\cdot)$ is only an approximate Bayes estimator, best for priors for which θ is near 1 with high probability (see Cressie, 1979, for similar considerations).

certain smoothness properties. For example consider the binomial example of subsection 3.1; it is easy to see that regardless of the prior G , $\delta_G(0) \leq \delta_G(1) \leq \dots \leq \delta_G(k)$. Hence any estimator $\hat{\delta}_G(\cdot)$ should reflect this. The monotonicity problem for discrete distributions has been solved by Van Houwelingen (1977). He has found a method that makes simple empirical Bayes estimators monotonic, but so that the risk of the estimator is never increased.

All of the families considered in this paper depended on only one parameter. In the case of multiparametric families, the identity (2.4) can still be used, but its usefulness may be limited. This is because parameters may not appear together in such a way that for each parameter, a functional R_0 can be found that satisfies (2.6). Nevertheless the importance of the one parameter discrete families fully justifies the investigation described herein.

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